



Conjugate heat transfer analysis of rectangular cooling channels using modeled and direct numerical simulation of turbulence

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ABSTRACT

We carry out a computational study of conjugate heat transfer in rectangular cooling channels typical of liquid propellant rocket engines, with the goal of establishing the predictive capability of Reynolds averaged Navier-Stokes (RANS) solvers. A classical Spalart-Allmaras RANS solver is here used, which is iteratively coupled with a Fourier solver which solves for conduction within the duct walls. Comparison is made with the reference solutions of the same problem obtained with a direct numerical simulation (DNS) approach, which allows to describe the full features of flow and heat evolution in the channel. Numerical simulations are carried out for different values of the fluid-solid thermal conductivity ratio to bring out conjugate heat transfer effects. Finite conductivity of the solid is found to imply reduced thermal efficiency of the overall system as a result of both increased thermal resistance, and asymmetric heat loading on the fluid. We find that, despite the absence of the secondary motions, the RANS/Fourier solver can accurately estimate the pressure drop. Differences in the prediction of thermal effects are generally larger. However, the wall temperature distributions are predicted by RANS with reasonable accuracy, and discrepancy in the overall heat transfer coefficient is no larger than 10%.

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1. Introduction

Challenges in the design of cooling systems for liquid rocket engines are unique because of the harsh environment in terms of flow and material temperatures, heat flux level and flow pressure [1]. Despite the acquired experience, the technological advancements for the realization of rocket thrust chambers allow the selection of innovative designs and materials as well as new liquid propellants as coolants. One of the keys to the success of a new technology is the capability to realize safe and efficient cooling systems, requiring adequate physical understanding of the phenomena and effects associated with the selection of a given material or design. In this respect, crucial support is provided by numerical simulations. The temperature distribution within the solid walls of a cooling jacket and the resulting heat fluxes strongly affect the system performance and material resistance, especially if reusable systems are envisaged. Moreover, the wall temperature is strictly dependent on the thermal conductivity of the wall materials and on the channel geometry. The resulting coupling between the temperature field inside the walls and the heat exchange capabilities of the coolant flow makes the numerical solution of the coolant

flow alone only partially effective. A conjugate heat transfer (CHT) approach is therefore needed to obtain the correct estimation of the solid wall and coolant flow temperature fields which includes different zones (solid wall and coolant flow) governed by different equation systems. In this manner, the estimation is physically consistent as a consequence of the coupling between the boundary conditions at the interfaces.

Several CFD solutions of the conjugate heat transfer problem in liquid rocket engines can be found in the literature. Pizzarelli et al. [2] addressed the problem of conjugate heat transfer for a single heated channel cooled by supercritical methane. The approach was based on a Reynolds averaged Navier-Stokes (RANS) solver for the supercritical fluid flow coupled with a Fourier equation solver. Results were relevant to a rectangular tube heated from one side which was tested experimentally to get more information about the behavior of the coolant in the given high-pressure/low-temperature conditions. The simulations showed good agreement and clear validation of the approach, nevertheless some uncertainty in the experiment, as for instance unmeasured heat loss, did not allow a full understanding of the limits and capabilities of the approach.

A thermally coupled simulation (conjugate heat transfer) of a fatigue experiment has been carried out by Barfusz et al. [3]. The aim was to obtain an accurate lifetime prediction and to increase

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Nomenclature

Nu	fluid Nusselt number
A	channel cross section area
c_f	skin friction coefficient
c_p	heat capacity
D_h	hydraulic diameter of the channel
f_D	Darcy-Weisbach friction factor
G	mass flow rate per unit area
h	convective heat transfer coefficient
k	thermal conductivity
L	channel length
N	number of cells
P	perimeter
p	pressure
p_0	total pressure
q	heat flux
s	curvilinear abscissa along the cross section fluid/solid interface
T	temperature
t	height (length in z-direction)
u	generic velocity component (u_i), axial component of velocity
w	width (length in y-direction)
x, y, z	Cartesian coordinates
y^+	dimensionless wall distance
Pr	fluid Prandtl number
Re	Reynolds number, based on D_h and bulk velocity
α	thermal diffusivity
η	nondimensional outward wall-normal coordinate
μ	dynamic viscosity
ν	kinematic viscosity
ρ	fluid density
τ	viscous stress
θ	nondimensional temperature
b	bulk
c	channel
e	exit
f	fluid
g	hot gas side (heated wall)
i	inlet
j	jacket, jacket solid-fluid interface
l	liner, liner solid-fluid interface
r	rib, rib solid-fluid interface
s	solid
w	wall
$\overline{(\quad)}$	space averaged quantity
$*$	dimensionless quantity

the lifetime of a liquid rocket engine thrust chamber, looking for more reliable data than obtained from Nusselt number correlations or experimental data fitting. The authors presented simulations containing a thermal FEM model of the fatigue specimen structure, RANS simulations of nine cooling channels and a Flamelet-based RANS simulation of the hot gas. The model showed good agreement with the temperature distribution and transients of the experiment and was able to predict the transient temperature fields of the structure. This allowed to use the conjugate heat transfer model as a base for sequentially coupled lifetime predictions in different configurations, e.g. for cooling channel design optimization.

Negishi et al. [4] and Daimon et al. [5] demonstrated the engineering capabilities of a RANS-based CHT analysis. The first set of simulations [4] addressed supercritical parahydrogen flows in the cooling channel of the sub-scale hydrogen-cooled thrust

chamber with regenerative cooling. Generally, the computed results compared well with the measured data, while some quantitative discrepancy with the experimental data was found for the wall temperature in the region of maximum heat flux. Fully three-dimensional simulations were carried out emphasizing the role of the secondary flow generated by channel curvature on the overall heat transfer capabilities of the coolant. A more recent full three-dimensional conjugate combustion and heat transfer simulation [5] addressed validation against experimental data for a water-cooled methane-oxygen calorimetric chamber.

The common aspect of the above-mentioned simulations is that they rely on validation through comparison with available experimental data, indicating that reasonable engineering estimates of the overall coolant capabilities may be achieved (or not) by the selected models. This type of validation is, however, not very useful to establish the full predictive capabilities of RANS models, which goes well beyond those of basic, empirical-correlation-based, 1D models. In order to assess the local behavior of RANS models and evaluate their predictive capabilities, local information is needed, which is not usually available from experiments. In order to fill in this gap, in this paper we carry out dedicated Direct Numerical Simulations (DNS) of conjugate heat transfer in a cooling channel with a rectangular cross section, under the assumption of fully developed flow. DNS has the advantage of yielding easy insight into all the flow properties, and thus it can be used as a benchmark for turbulence models. Although DNS of conjugate heat transfer have previously been carried out in simplified geometries such as planar channels [6,7], we are not aware of any DNS of CHT in ducts. The DNS data will be used to evaluate the predictive capabilities of a CHT solver for spatially developing duct flow based on the coupling between a RANS module for the fluid and a Fourier solver for the fluid phase. The RANS closure under scrutiny is the one-equation model of Spalart and Allmaras which is often used as a trade-off between simplicity and predictive capability. RANS simulations are conducted by a solver which was previously validated against experimental data for supercritical pressure coolant in a heated channel [2,8].

The paper is organized as follows. In Section 2 we illustrate the set-up for the CHT test cases; in Sections 3 and 4 we provide details about the RANS and the DNS solvers, respectively; results are presented in Section 5, and concluding remarks are given in Section 6.

2. Definition of test cases

We study CHT in a straight cooling channel with rectangular cross section and smooth walls, subjected to a uniform heat flux at its lower side, herein referred to as liner (Fig. 1). In order to facilitate comparison of numerical results, the coolant is assumed to be an incompressible fluid with uniform and isotropic properties (density, viscosity, specific heat, and thermal conductivity), hence the relationship between heat flux and the temperature difference between wall and bulk temperature is linear. The solid wall is also assumed as a material with uniform and isotropic properties (conductivity, specific heat, density). As a result, both the heat transfer coefficient and the pressure gradient are independent of the input heat flux, once the flow is fully developed.

The main geometrical properties describing the channel cross section are given in Fig. 2, in which the hydraulic diameter is assumed as the reference length. The channel length is assumed to be either $240 D_h$ or $480 D_h$ in the (space-developing) RANS simulations, and $6 D_h$ in DNS, in which a fully developed flow is assumed on account of streamwise periodicity. Note that, despite the general validity of the present analysis for heated rectangular channels, we prefer to follow the nomenclature of rocket cooling channels because of the framework of the present research.

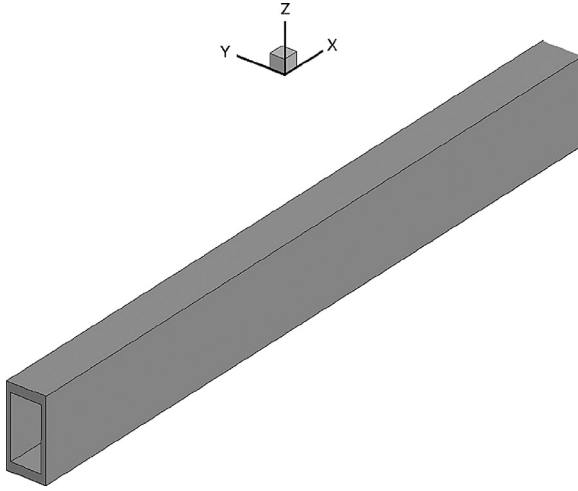


Fig. 1. Sketch of the cooling channel geometry.

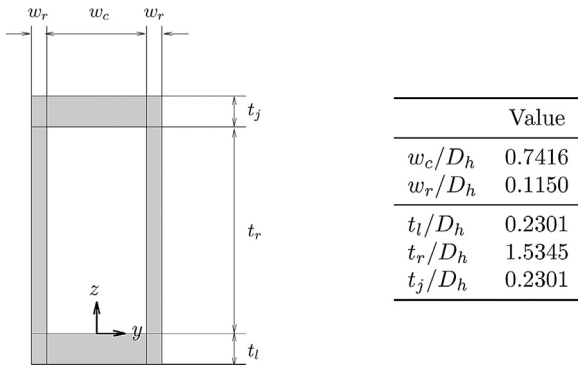


Fig. 2. Schematic of the channel cross section, with definition of the main geometrical parameters.

The bulk Reynolds number, based on the mean flow velocity u_b ($= 1/A \int_A u dA$), the hydraulic diameter D_h , and the fluid kinematic viscosity is $Re_b = 22440$, at which DNS is feasible, and which is not far from the range of applicative interest. RANS and DNS solutions are hereafter mainly compared in terms of their friction and heat transfer coefficients in the fully developed region. For that purpose, the Darcy-Weisbach friction factor (f_D) is evaluated from the computed streamwise pressure drop, according to

$$\frac{dp_0}{dx} = -f_D \frac{\rho u_b^2}{2D_h}, \quad (1)$$

and numerical results are compared with the correlation for rectangular ducts proposed by Shah and Sekulib [9],

$$f_D = 0.00512 + 0.4572 Re_{b1}^{-0.311}, \quad (2)$$

where Re_{b1} is based on a modified hydraulic diameter,

$$D_1 = D_h [0.667 + 0.458 \alpha^* (2 - \alpha^*)], \quad (3)$$

with α^* as the ratio of the short to long wall lengths. In the present test case, $\alpha^* = t_r/w_c = 0.483$, thus $D_1 = 1.002 D_h$, and at the selected value of Re_b we get $f_D = 0.0254$, which only slightly differs from Colebrook's equation for smooth circular pipes ($f_D = 0.0252$).

Comparison of the results is made also locally through the local skin friction coefficient, defined in terms of the local wall shear stress, τ_w

$$c_f = \frac{2\tau_w}{\rho u_b^2}, \quad (4)$$

Table 1

List of flow parameters for the computed flow cases. Re_b is the bulk Reynolds number, Pr is the fluid Prandtl number, and the subscripts s and f refer to properties of the fluid and of the solid, respectively.

Flow case	Re_b	Pr	$(\rho_s c_{p,s})/(\rho_f c_{p,f})$	k_s/k_f
ISOTH	22,440	6.033	1.000	∞
HIGHCOND	22,440	6.033	1.000	164
LOWCOND	22,440	6.033	1.000	16.4

and of its average values along the duct perimeter ($\bar{c}_f$), and along each side (liner, rib, and jacket walls), which are defined as follows

$$\bar{c}_{f,l} = \frac{2\bar{\tau}_{w,l}}{\rho u_b^2}; \quad \bar{c}_{f,r} = \frac{2\bar{\tau}_{w,r}}{\rho u_b^2}; \quad \bar{c}_{f,j} = \frac{2\bar{\tau}_{w,j}}{\rho u_b^2}, \quad (5)$$

where

$$\bar{\tau}_{w,l} = \frac{1}{w_c} \int_{-w_c/2}^{w_c/2} \tau_{w,l} dy, \quad \bar{\tau}_{w,r} = \frac{1}{t_r} \int_0^{t_r} \tau_{w,r} dz, \\ \bar{\tau}_{w,j} = \frac{1}{w_c} \int_{-w_c/2}^{w_c/2} \tau_{w,j} dy. \quad (6)$$

The temperature distribution within the fluid and the solid is also a function of the fluid Prandtl number ($Pr = \mu c_{p,f}/k_f$), and of the relative thermal properties of the solid and the fluid, for instance expressed in terms of the ratio of the thermal conductivities (k_s/k_f), and of the volumetric heat capacities ($(\rho_s c_{p,s})/(\rho_f c_{p,f})$). The fluid Prandtl number is assumed to be $Pr = 6.033$ for all simulations as representative of water. The ratio of the solid and fluid volumetric heat capacities is assumed to be unity because in the present study the focus is on the heat transfer process at fully developed conditions rather than on the energy storage. The ratio is also selected according to experience that shows that for fluids and solids of interest for the present applications temperature fluctuations are quite small. Different values of the thermal conductivity ratio have been considered, starting from the reference case in which the solid is artificially kept at uniform temperature, as would be the case if $k_s/k_f \rightarrow \infty$, and referred to as ISOTH case. Two test cases with finite conductivity ratios have further been considered to study coupling effects, namely $k_s/k_f = 164$ (referred to as HIGHCOND), and $k_s/k_f = 16.4$ (referred to as LOWCOND). The LOWCOND thermal conductivity value is representative of an Inconel alloy at low temperature, whereas for the HIGHCOND a value ten times higher is selected for comparison. A list of the relevant parameters for the three test cases is provided in Table 1.

Average heat fluxes are obtained by integrating the local heat flux q_w over the whole cross section perimeter ($\bar{q}_{w,c}$) and also by considering separately the liner wall ($\bar{q}_{w,l}$), the rib wall ($\bar{q}_{w,r}$) and the jacket wall ($\bar{q}_{w,j}$), as follows

$$\bar{q}_{w,c} = \frac{1}{P} \int_P q_w ds, \quad \bar{q}_{w,l} = \frac{1}{w_c} \int_{-w_c/2}^{w_c/2} q_w dy, \\ \bar{q}_{w,r} = \frac{1}{t_r} \int_0^{t_r} q_w dz, \quad \bar{q}_{w,j} = \frac{1}{w_c} \int_{-w_c/2}^{w_c/2} q_w dy. \quad (7)$$

Similarly, we define average wall temperatures as follows,

$$\bar{T}_{w,c} = \frac{1}{P} \int_P T_w ds, \quad \bar{T}_{w,l} = \frac{1}{w_c} \int_{-w_c/2}^{w_c/2} T_w dy, \\ \bar{T}_{w,r} = \frac{1}{t_r} \int_0^{t_r} T_w dz, \quad \bar{T}_{w,j} = \frac{1}{w_c} \int_{-w_c/2}^{w_c/2} T_w dy, \quad (8)$$

and we define the associated heat transfer coefficients,

$$h_c = \frac{\bar{q}_{w,c}}{\bar{T}_{w,c} - T_b}, \quad h_l = \frac{\bar{q}_{w,l}}{\bar{T}_{w,l} - T_b},$$

$$h_r = \frac{\bar{q}_{w,r}}{\bar{T}_{w,r} - T_b}, \quad h_j = \frac{\bar{q}_{w,j}}{\bar{T}_{w,j} - T_b}. \quad (9)$$

In Eq. 9, T_b is the coolant bulk temperature,

$$T_b = \int_A T \rho u dA / \int_A \rho u dA, \quad (10)$$

which for the case of constant property has the meaning of the mass-averaged total enthalpy. The driving force for the temperature field is the difference between T_b and the average wall temperature at the heated side of the liner, $T_{w,g}$,

$$\bar{T}_{w,g} = \frac{1}{w_c + 2w_r} \int_{-w_c/2-w_r}^{w_c/2+w_r} T_{w,g} dy \quad (11)$$

where the corresponding average heat flux is

$$\bar{q}_{w,g} = \frac{1}{w_c + 2w_r} \int_{-w_c/2-w_r}^{w_c/2+w_r} q_w dy. \quad (12)$$

For the purpose of reporting the numerical results, it is convenient to use suitably normalized velocity and temperature distributions. With regard to the velocity field, we use the bulk velocity for normalization and define

$$u^* = \frac{u}{u_b}. \quad (13)$$

Regarding the temperature field, we define a nondimensional temperature θ as

$$\theta = \frac{T - T_b}{\bar{T}_{w,g} - T_b}, \quad (14)$$

in such a way that its average values are uniform under thermally developed flow conditions, with $\bar{\theta}_{w,g} = 1$, and $\theta_b = 0$, by construction.

The heat transfer performance of the cooling channel is evaluated in terms of the fluid Nusselt number, based on the above defined heat transfer coefficient (see Eq. 9),

$$\overline{Nu}_c = \frac{h_c D_h}{k_f}, \quad (15)$$

or in nondimensional variables,

$$\overline{Nu}_c = \left(\frac{\partial \theta}{\partial \eta} \right)_{w,c} \frac{1}{\bar{\theta}_{w,c}}, \quad (16)$$

where η is the outward wall-normal coordinate nondimensionalized with D_h . A reference correlation frequently used for estimating the mean heat transfer coefficient of cooling channels is the Gnielinski analogy [9,10], according to which

$$\overline{Nu}_c = \frac{(f_D/8)(Re - 1000)Pr}{1 + 12.7(f_D/8)^{1/2}(Pr^{2/3} - 1)}, \quad (17)$$

which is to be coupled with Eq. (2), yielding f_D . According to Shah and Sekulib [9], this expression applies within $\pm 9\%$ uncertainty for smooth ducts with uniform heating. However, if heating the duct perimeter is nonuniform, as in the present case, the predictions are quoted to be up to 20% higher than measured values. For the present test case, on the basis of the above mentioned values of f_D and Re , the resulting value obtained from Eq. 17, is $Nu = 155.0$. The global heat transfer performance of the fluid+solid system can also be quantified in terms of the Nusselt number based on the heat flux and mean temperature at the hot side of the liner, namely

$$\overline{Nu}_g = \frac{\bar{q}_{w,g} D_h}{k_f (\bar{T}_{w,g} - T_b)}, \quad (18)$$

or in nondimensional variables, being the total heat per unit length entering the cross section $\bar{q}_{w,g}(w_c + 2w_r) = \bar{q}_{w,c}P$,

$$\overline{Nu}_g = \left(\frac{\partial \theta}{\partial \eta} \right)_{w,g} = \overline{Nu}_c \frac{P}{w_c + 2w_r} \bar{\theta}_{w,c}. \quad (19)$$

The local heat transfer performance along the cooling duct walls can also be evaluated in terms of the locally defined Nusselt number,

$$Nu = \left(\frac{\partial \theta}{\partial \eta} \right)_w \frac{1}{\bar{\theta}_w}. \quad (20)$$

For later use we note that under fully developed conditions, upon neglect of heat transfer through the duct ribs and under the assumption of $w_r \ll w_c$, the following relationship between the mean Nusselt number and the mean temperature at the cold side of the liner holds,

$$\bar{T}_{w,l} - T_b = \frac{1}{1 + \beta} (\bar{T}_{w,g} - T_b), \quad (21)$$

or in nondimensional terms,

$$\bar{\theta}_{w,l} = \frac{1}{1 + \beta}, \quad (22)$$

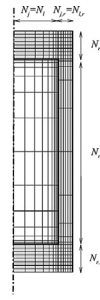
where $\beta = \overline{Nu}_c (k_f/k_s) (t_l/D_h)$.

3. RANS Setup

A conjugate heat transfer model based on the coupled numerical integration of the Navier-Stokes equations for the coolant flow and the Fourier's law of conduction for the heat transfer within the wall is adopted in the present study [8,11]. In particular, the coolant flow solver integrates the Reynolds Averaged Navier-Stokes Equations, written in the conservation form, by a Godunov-type finite volume scheme, which is second-order accurate in space. Its main feature is the capability to deal with any compressible fluid whose behavior is represented by a generic equation of state, as the fluid thermodynamic properties are stored in a look-up table.

Turbulence is computed according to the one-equation model of Spalart-Allmaras [12]. Validation of the coolant flow solver has been reported by comparison made with available experimental data on supercritical nitrogen [13] and transcritical hydrogen [14]. The coupled solution of the coolant flow and the wall heat conduction is achieved by an iterative procedure in which the wall temperature distribution obtained by a finite-volume solver, which is second-order accurate in space, of the Fourier's thermal problem is used as boundary condition for the coolant flow computations. Validation of the conjugate heat transfer approach for a cooling channel with rectangular cross section and asymmetric heating has been reported by Pizzarelli et al. [2]. In the present study, the simulation of the test case presented in Section 2 is carried out considering constant thermodynamic properties of the fluid.

The numerical analysis is conducted on fluid and solid domains that reproduce the channel geometric features described in Fig. 2. As for the boundary conditions, the imposed inflow conditions are the static temperature T and the mass flow rate per unit area G , while the outflow condition is obtained by prescribing the value of the static pressure p . The wall boundary condition of incoming heat flux q_w is imposed on the bottom of the solid wall block. Adiabatic boundary conditions are considered on all other wall surfaces (side, top, front, back). Symmetry conditions are finally enforced for the wall and fluid computation at the symmetry plane. In fact, for the prescribed boundary conditions (heating from the bottom and adiabatic walls at the side and at the top) and because of the given geometry with rectangular cross section, see Fig. 2, the steady-state flow and temperature solution must be symmetrical with respect to the vertical plane at the middle of the channel. The mesh topology (Fig. 3) exploits the symmetry to save computational cells and clustering of nodes in the flow domain to resolve the boundary layer. Note that the solid domain is organized in five rectangular blocks surrounding the fluid domain. The first block is placed below the channel and is wide one half the width of the



	Flow			Wall						
	N_x	N_y	N_z	N_x	N_y	$N_{x,r}$	$N_{y,l}$	N_r	$N_{r,j}$	
Grid A (coarse)	72	12	16	72	8	6	8	10	8	
Grid B (reference)	144	24	32	144	16	12	16	20	16	
Grid C (fine)	288	48	64	288	32	24	32	40	32	

Fig. 3. Channel cross section (Grid A) and number of cells for the three grids used for RANS simulation in the grid refinement study ($L = 240 D_h$).

channel $w_c/2$ and high as the liner t_l . The second block is on his right (see Fig. 3) and is wide w_r and high t_k . The third block is above the second and to the right of the fluid domain: its width is w_r and its height is t_r . The fourth block is above the third one: its width is w_r and its height is t_j . The fifth block is to the left of the fourth and above of the fluid domain: its width is $w_c/2$ and its height is t_j .

Turbulence model parameters are the standard values of the Spalart-Allmaras values with the turbulent Prandtl number $Pr_t = 0.9$.

3.1. Grid convergence

A grid refinement study for the test case geometry has been carried out. The assumed channel length is $L/D_h = 240$. The number of cells of the coarse, reference, and fine grids are reported in Fig. 3, indicating the streamwise, horizontal and vertical grid resolution in the three spatial directions, respectively, for both the flow and wall regions. The resolution of the different grids is obtained while keeping the same clustering laws.

Therefore, the resulting values of the dimensionless wall distance y^+ changes according to the grid level. Except the inlet region, the maximum value of y^+ of a wall cell occurring for each grid is about 3.5, 1.0, and 0.5 for the coarse, reference, and fine grids, respectively.

Values of order one, which are required to solve up to the wall, are thus obtained for all grids except the coarsest one. Note that the maximum value of y^+ is significantly higher than the average on the bottom and side walls for each of the solved grids. A first grid convergence analysis has been carried out on the main bulk properties: temperature and pressure. In Figs. 4(a), and 4(b) the evolution of the bulk temperature and the coolant pressure along the channel is reported for the three grids. Values are nondimensionalized with the inlet temperature T_i and with the exit pressure p_e , respectively. Fig. 4(a) shows that no effect of the mesh on the result of bulk temperature is detected. This is an expected result indicating that the energy balance is correctly satisfied, independently of resolution.

On the other hand, the results reported in Fig. 4(b) for pressure show a visible displacement at the coarsest levels. However, the changes between reference and fine grid are quite smaller, such that the adoption of the fine mesh in place of the reference one is not justified. Overall, the reference grid features on the one hand a good resolution at the wall to allow for solving the boundary layer up to the wall (i.e., y^+ order one) and on the other hand the final solution quite close to that of the fine grid. It is for this reason

that Grid B has been selected as the reference grid for the studies carried out in Section 5.

A further analysis of grid convergence has been done on the evaluation of overall friction (Fig. 5(a)) and heat transfer coefficients (Fig. 5(b)). Friction factor and Nusselt number are presented as their ratio to the corresponding theoretical (empirical) value obtained by Eqs. 2 and 17, respectively.

The friction factor behavior is based on the derivative of pressure and therefore it clearly follows what can be seen in Fig. 4(b). Fig. 5(a) allows to appreciate an almost second-order convergence of the value of f_D which quickly reaches its asymptotic value. Note that the asymptotic value, starting from a value higher than the one from the correlation, converges towards a value clearly lower than theoretical. The final value obtained by Richardson extrapolation [15] is also reported in Fig. 5(a) and corresponds to $f_D = 0.0240$.

The Nusselt number shows a similar behavior, quickly reaching the asymptotic value. Its behavior in the ISOTH case shows a second-order behavior Fig. 5(b). The final value obtained by Richardson extrapolation [15] is also reported in Fig. 5(b) as the extrapolated dashed line and corresponds to $Nu = 135.6$. Note that the asymptotic value of the Nusselt number is significantly lower than the one of the correlation, and this shows the importance to discuss the present results with DNS solutions as will be done in Section 5. A similar behavior is found for the CHT cases, with a slightly lower order of convergence due to the coupling between RANS and Fourier solvers and to the longer channels required to reach the fully developed regime.

3.2. Flow development

Despite the results of the grid convergence analysis show that the main global parameters of interest, i.e friction factor and Nusselt number reach their asymptotic values within about 50 and 100 hydraulic diameters from the entrance, indicating that fully developed flow conditions are reached, a further analysis is carried out to verify that developed flow condition are reached locally. To this aim, the nondimensional velocity and temperature profiles are analyzed in the following. The analysis of the velocity profiles for Grid A allows to verify the hydraulic flow development along the channel.

In Fig. 6, the velocity profiles are shown on the vertical and horizontal geometrical symmetry planes ($z = t_r/2$ in the top panel and $y = 0$ in the bottom panel). It is found that the velocity profiles are practically unaltered at axial positions downstream of $x \cong 50 D_h$. Both at the planes of half width and half height of the channel, it is clearly seen that the velocity profile is unchanged between 120 and 240 hydraulic diameters from the entrance. The solution therefore shows a fully developed velocity profile at the end of the channel.

As we have seen in the above discussion, the behavior of temperature and velocity parameters are similar in the ISOTH case. We have also anticipated that asymmetrically heated CHT simulations require a longer length to reach full thermal development. The behavior of the temperature development is therefore shown for the LOWCOND case to highlight the delayed temperature field development. The analysis is therefore carried out for a longer channel. In particular, Fig. 7 is obtained with the same number of cells at the cross section as Grid B but twice the number of cells in x-direction and twice the length of the channel.

The results are shown also for temperature as profiles on the vertical and horizontal geometrical symmetry planes ($z = t_r/2$ in the top panel and $y = 0$ in the bottom panel). It is interesting to highlight that there are still significant displacements between 240 and 480 diameters. Another interesting feature is that the temperature profile at half height of the channel quickly reaches full de-

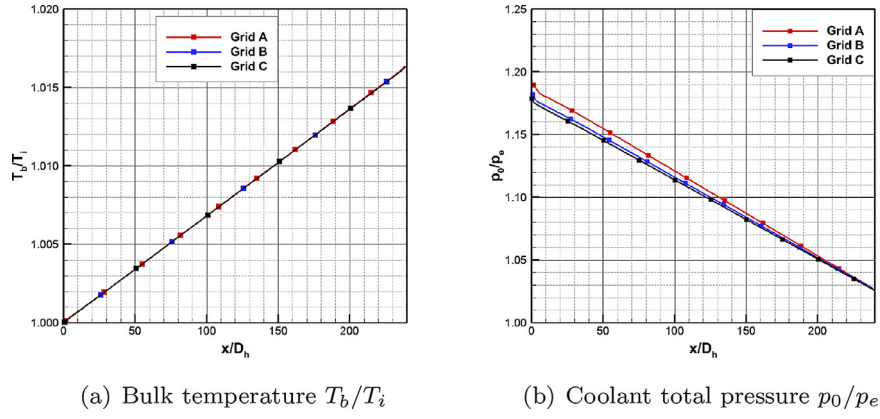


Fig. 4. Grid convergence analysis for bulk temperature and pressure evolution along the channel.

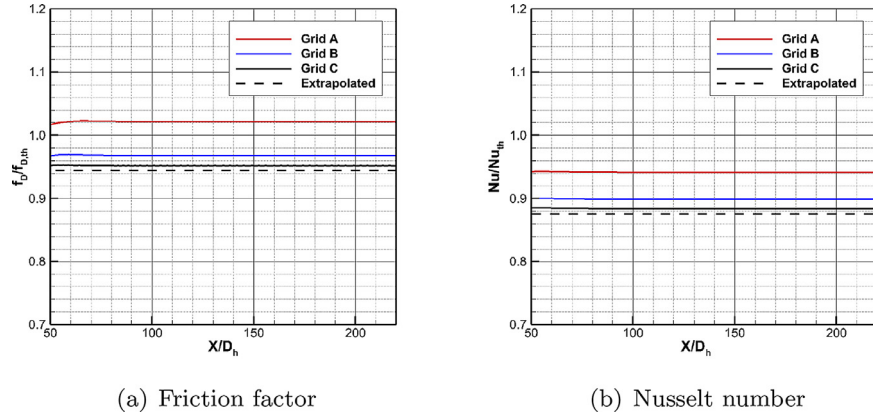


Fig. 5. Grid convergence analysis for the friction and heat transfer parameters normalized with respect to correlation values.

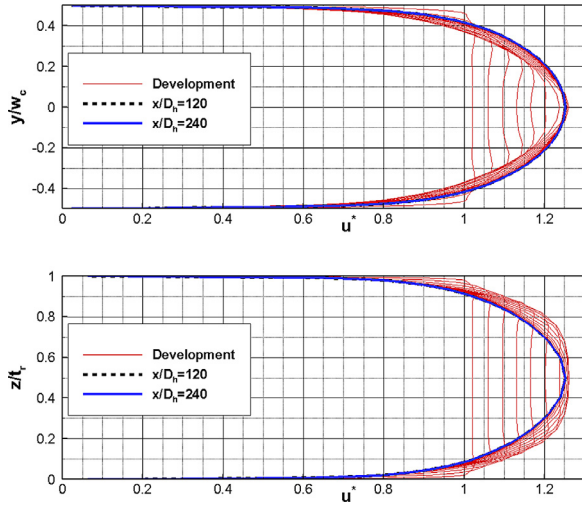


Fig. 6. Nondimensional velocity profiles (the x -component, u^*) as a function of y and z at different axial positions, on the two channel symmetry planes.

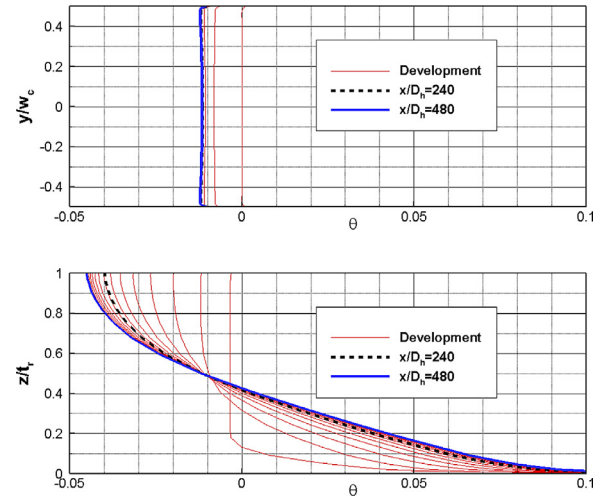


Fig. 7. Nondimensional temperature profiles (θ) as a function of nondimensional y and z coordinates at different axial positions, on the two channel symmetry planes.

development (top panel). On the other hand, a slower evolution is seen at both the liner and jacket walls (bottom figure). From the liner to half rib height, a channel long 240 diameters seems still to be sufficient to reach fully developed conditions, whereas, from half rib height to jacket, the development is definitely slower and the need of the longer channel is highlighted by the visible difference between temperatures at 240 and 480 diameters. However, at

the end of this longer channel, the asymptotic value is eventually reached.

4. DNS Setup

The reference solution for fully developed flow is computed by means of DNS, hence simulation of the Navier-Stokes equations in their unsteady, three-dimensional form, with sufficient resolution to resolve all the scales of turbulent flow motion, down to the

dissipative (Kolmogorov) scale. Assuming an incompressible, ideal fluid model, we solve for

$$\frac{\partial u_j}{\partial x_j} = 0, \quad \frac{\partial u_i}{\partial t} + \frac{\partial u_i u_j}{\partial x_j} = -\frac{\partial p}{\partial x_i} + \nu \frac{\partial^2 u_i}{\partial x_j \partial x_j} - \frac{d\bar{p}}{dx} \delta_{i1}, \quad (23)$$

where $d\bar{p}/dx$ is the pressure gradient required to balance friction and maintain a constant mass flow rate along the streamwise direction. The equations are discretized in a Cartesian coordinate system using staggered central second-order finite-difference approximations, to guarantee that the kinetic energy is globally conserved in the limit of inviscid flow [16]. Time advancement is carried out by means of a hybrid third-order low-storage Runge-Kutta algorithm coupled with the second-order Crank-Nicolson scheme combined in the fractional-step procedure, whereby the convective terms are treated explicitly and the diffusive terms are treated implicitly in the cross-stream directions (y and z). The Poisson equation for the pressure field stemming from the incompressibility condition is efficiently solved through Fourier transform-based methods [17]. The fluid equations are augmented with the transport equations for the temperature field

$$\frac{\partial T}{\partial t} + \frac{\partial T u_j}{\partial x_j} = \frac{1}{\rho c_p} \frac{\partial}{\partial x_j} \left(k \frac{\partial T}{\partial x_j} \right) + Q, \quad (24)$$

where ρc_p and k are the local volumetric heat capacity and thermal conductivity, and Q is a suitable heating term, which is dynamically adjusted in such a way that the bulk temperature (T_b) across the channel is kept constant in time. In the numerical discretization, the transported variable is located at the cell center, in such a way that the variance of T is exactly preserved in the limit of inviscid flow. Periodic boundary conditions are used in the axial direction for both the velocity and temperature fields, as appropriate for fully developed flow conditions. Extensive validation of the solver was carried out in previous studies [18,19]. In particular, in the work of Pirozzoli et al. [19] we carried out DNS of turbulent flow in a duct with square cross section up to $Re = 40000$, focusing on the structure of the secondary motions.

To study conjugate heat transfer problems, the solver has been extended, by still solving for Eqs. (23), (24) at all grid points, but forcing $u_j = 0$ at solid points, and adjusting the thermal properties within the fluid and the solid phase. The numerical simulation has been carried out in a box with identical cross-stream geometry as for RANS, and with length $L_x/D_h = 6$. A mesh consisting of $768 \times 768 \times 384$ grid points has been used, so as to satisfy the traditionally accepted criterion that the mesh spacing in the wall-normal direction should be less than one wall unit at the wall, and in the cross-stream directions it should be less than ten wall units [20]. The simulations have been advanced in time for about four thousand time units (based on D_h/u_b), on account of the extreme disparity between the fast time scales of turbulence and the long time scales associated with thermal diffusion within the solid. Flow statistics have been collected by averaging in the axial direction and in time over an interval of about $1000D_h/u_b$, to achieve proper convergence of the mean flow statistics, including friction and heat transfer coefficients, as well as convergence of the secondary motions.

5. Results

The results achieved with spatially developing RANS simulations in a $480 D_h$ long channel and with DNS for fully developed flow are hereafter compared, for the three cases listed in Table 1. Comparison of the DNS flow statistics with the RANS simulations is carried out at exit section, at which the flow is very nearly fully developed.

Table 2

Comparison of friction factor obtained with RANS, DNS and correlation (2).

	RANS	DNS	Eq. (2)
f_D	0.0239	0.0243	0.0254
difference	-1.6%	-	+4.5

5.1. Velocity field

The first comparison addresses the evaluation of velocity field and of the associated friction coefficient, as obtained from simulations and from established empirical relations. The analysis of friction is not repeated for all the flow cases because, owing to the assumption of constant fluid and solid properties, the momentum equation is fully uncoupled from the temperature field. Hence, the following data, taken from the LOWCOND flow case are identical to those obtained for the ISOTH and HIGHCOND cases.

Maps of the instantaneous axial velocity field in a cross-stream plane computed with DNS are shown in Fig. 8, at three representative time instants.

For the sake of interpretation, the blue zone surrounding the channel marks the solid domain, where the velocity is zero. As expected, the velocity field exhibits significant deviations from the averaged state, with the presence of low-speed fluid intrusions from the wall to the core region, and conversely high-speed fluid sweeps toward the wall.

The axial velocity distributions from RANS (obtained with grid B) and DNS are compared in Fig. 9. In both cases, the maximum velocity exceeds the bulk value by about 25%. However, the contour lines away from walls tend to be more rounded in RANS, whereas in DNS they tend to more closely conform to the duct shape. As a result, the velocities near the duct corners tend to be higher in the DNS. This difference may be attributed to the inability of RANS based on the linear eddy-viscosity ansatz (and in particular, of the Spalart-Allmaras model) to predict secondary eddies, thus inhibiting additional momentum and heat transfer from the duct center toward the corners [21]. This interpretation is supported by the visualization of the mean streamlines in the channel cross section obtained in DNS, as shown in Fig. 10. Four pairs of asymmetric rollers are observed which very nearly satisfy the symmetry constraints imposed by the duct shape, and whose effect is bringing high-speed fluid from the core flow to the corners to compensate for momentum defect. The question is whether this large topological change translates to differences in terms of the friction and heat transfer coefficients of engineering interest. Indeed, it has been recently shown [19], that the effect of the secondary motions on the friction coefficient in a square duct is quite small, and quantifiable in terms of a few percent variation. Hence, while secondary motions yield a large change in the distribution of the friction coefficient along walls, their artificial suppression is not expected to yield substantial changes in the global flow properties.

Next, in Table 2 we compare the friction factors obtained from RANS and DNS with those predicted from the correlation (2). In the case of square ducts, the correlation (2) yields predictions of the friction coefficient which are accurate to within 0.2% of the DNS results [19]. Here, the 4.5% difference with respect to DNS is due to the rectangular shape of the cross section, which is not properly accounted for by the correlation. This is also in line with the reported uncertainty of about $\pm 5\%$ when the correlation (2) is applied to cross-sectional shapes other than circular [9]. Despite inability to capture the secondary motions, the friction coefficient obtained with RANS is only about 1.6% less than the DNS estimate, hence in line with expectations [18].

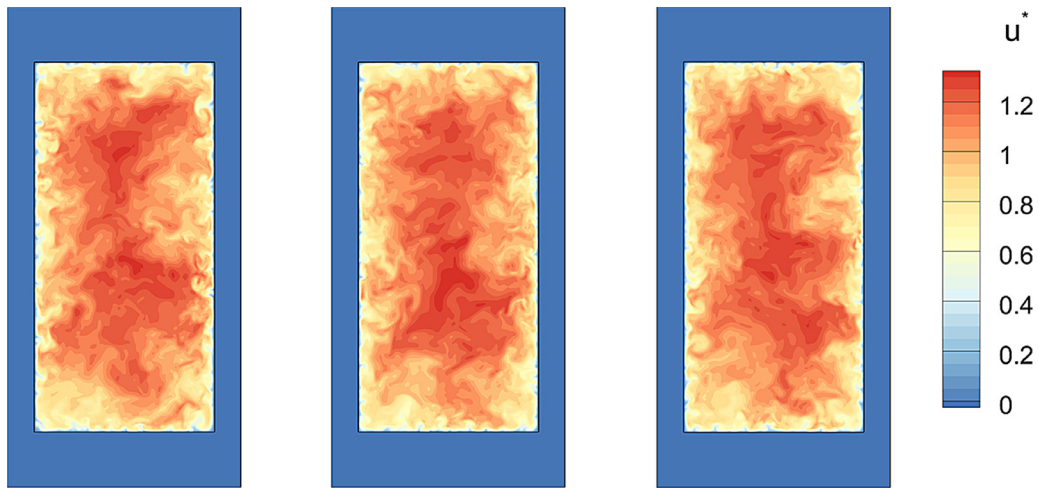


Fig. 8. Instantaneous nondimensional axial velocity distribution in a cross-stream plane obtained from DNS, at three representative time instants.

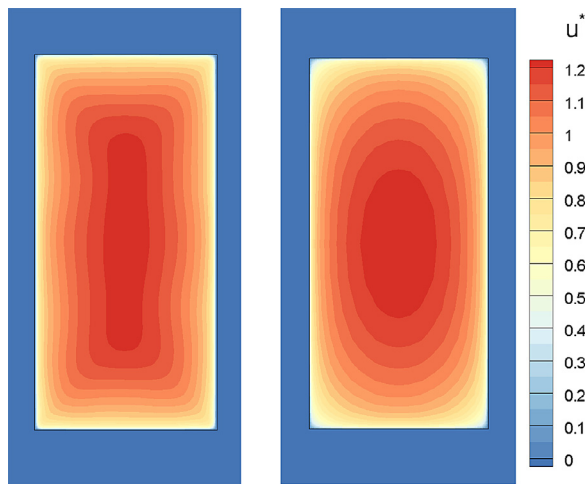


Fig. 9. Comparison of nondimensional axial velocity distributions (u^*), as obtained from time average of DNS (left) and at channel exit by RANS (right).

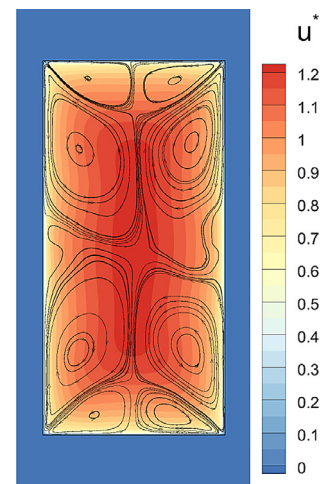


Fig. 10. Mean streamtraces in the channel cross section, as obtained from time average of DNS results.

A more detailed comparison is reported in Fig. 11, where we show the distributions of the local skin friction coefficient c_f , defined in Eq. (4), along the duct walls. In the figure we also report the reference mean value obtained from Eq. (2), namely $c_f = f_D/4 = 0.00634$. As expected from symmetry, the distributions on opposing walls are nearly identical, both in DNS and RANS. DNS

is characterized by a nearly flat distribution of c_f over each wall, with secondary peaks near the corners, where of course $c_f = 0$. Friction is slightly less than the average value provided by correlation on the short sides (liner and jacket), and a bit larger on the long sides (ribs). Hence, the overprediction of the friction correlation (dashed line) seems to be mainly due to friction drop toward

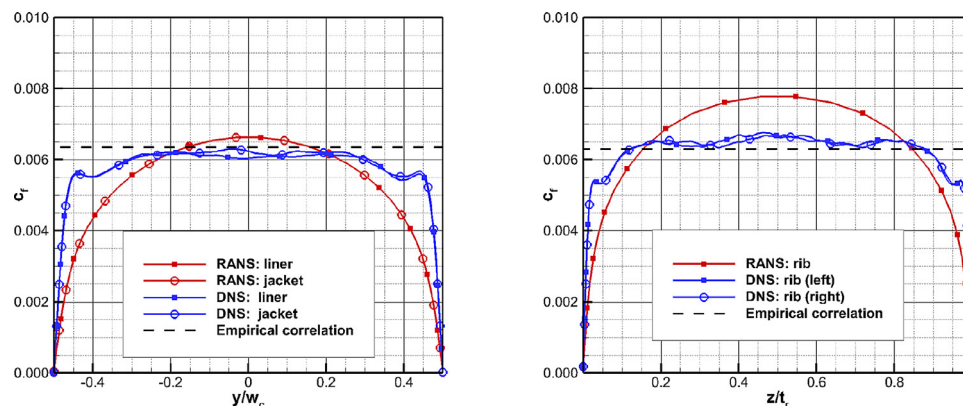


Fig. 11. Skin friction coefficient distribution along liner and jacket wall (left), and rib walls (right).

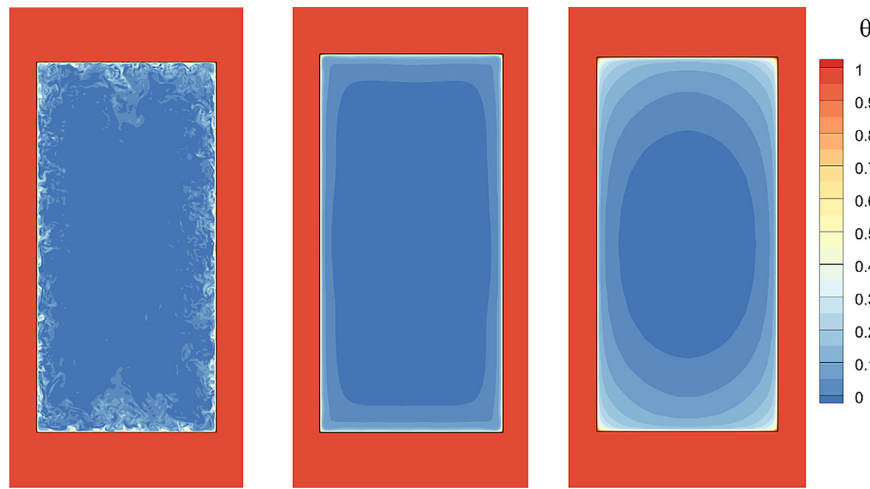


Fig. 12. ISOTH flow case: nondimensional temperature fields (θ) from DNS (instantaneous, left, and averaged, middle), and at the channel exit from RANS (right).

corners, which might have less impact at high Re_b . RANS is instead characterized by a roughly parabolic distribution, with no plateau. As in DNS, the average friction is higher on the long sides, where RANS clearly overshoots the DNS values. However, on account of underprediction on the short sides, the overall friction coefficient is similar.

5.2. Temperature field: Isothermal wall

The reference case for the analysis of the temperature field is the ISOTH flow case, in which the temperature within the solid is explicitly assigned, and which would correspond to a case in which the solid thermal conductivity is infinite.

The temperature fields obtained with DNS and RANS are shown in Fig. 12, which clearly shows that $\theta = 1$ throughout the solid. The instantaneous temperature field in DNS (left panel) clearly brings out the presence of plumes associated with turbulent transfer of heat from the walls to the core flow. The hint of convective heat transfer from secondary motions can also be inferred from the hotter columns rising from the side bisectors towards the core. The mean temperature distributions from DNS and RANS have the same qualitative structure as the velocity field (see Fig. 8), with contours in DNS following more closely the shape of the duct. Careful inspection, however, shows that the temperature gradients at the walls are higher than the velocity gradient, as the Prandtl number is significantly higher than unity, thus the thermal boundary layer is thinner than the kinematic boundary layer.

The global heat transfer characteristics of the duct are summarized in Table 3, where we report the computed Nusselt numbers, as defined in Eqs. (16) and (18), along with the predictions of Eq. (17). Although friction was somewhat overpredicted from Eq. (2), the heat transfer correlation returns a value of the Nusselt number which is nearly identical to DNS, most likely as a result of cancellation of opposing errors. On the other hand, the underprediction of the heat transfer coefficient for RANS is more significant than for friction, amounting to about 10%. Higher discrepancy is also evident in the distribution of the local Nusselt number, shown in Fig. 13. The qualitative behavior is similar to the skin friction, namely roughly parabolic for RANS and nearly flat for DNS. Comparison between Fig. 11 and Fig. 13 shows stronger near-wall peak of Nu in DNS as compared to c_f . Regarding RANS, the large underestimation of the heat transfer coefficient occurring at the short sides is not compensated at the long sides, unlike what observed for c_f . Reasons for the impaired similarity between friction and heat transfer coefficients may reside in the relatively high value

Table 3

Fluid ($\overline{Nu}_c$) and global ($\overline{Nu}_g$) Nusselt number obtained by RANS, DNS and from correlation (17), and percent differences with respect to DNS. The RANS values are obtained on Grid B.

Test case		RANS	DNS	Eq. 17
ISOTH	$\overline{Nu}_c$	138.9	154.6	155.0
	difference	-10.1%	-	-0.3%
	$\overline{Nu}_g$	651.6	724.5	-
	difference	-10.1%	-	-
HIGHCOND	$\overline{Nu}_c$	127.2	146.5	155.0
	difference	-13.0%	-	+5.8%
	$\overline{Nu}_g$	130.9	143.1	-
	difference	-8.5%	-	-
LOWCOND	$\overline{Nu}_c$	114.7	138.7	155.0
	difference	-17.1%	-	+11.8%
	$\overline{Nu}_g$	41.6	43.6	-
	difference	-4.6%	-	-

of the Prandtl number, which may entail a higher flow turbulent Prandtl number Pr_t than the standard value of 0.9 assumed in the present simulations [22].

Information about the detailed flow features is provided by DNS, as reported in Fig. 14, where we show the distributions of the local temperature fluctuations and of the fluctuating velocity and temperature wall-normal gradients in the plane of the liner wall and of the rib wall. As expected, the near-wall velocity field is organized into streaks with alternating low- and high-momentum, which are the key signature of wall turbulence [23]. As it should be, the wall temperature fluctuations are identically zero for this flow case, as isothermal wall boundary conditions are enforced. Interestingly, the fluctuating wall-normal temperature gradients at both walls show close similarity with the velocity gradient field, and a clear negative correlation is apparent, with high-speed velocity streaks associated with negative streaks of the local heat flux. This effect is well recognized in canonical flows such as boundary layers, channels, and pipes at Prandtl number around unity [24], being associated with the formal similarity of the streamwise momentum and temperature equations. However, to the best of our knowledge, this similarity is far from established in the case of (relatively) complex ducts, and/or for Pr significantly higher than unity as in the present case.

5.3. Temperature field: high conductivity ratio

It is worthwhile mentioning that, if on the one hand, the mean velocity distribution and the pressure gradient are not affected by

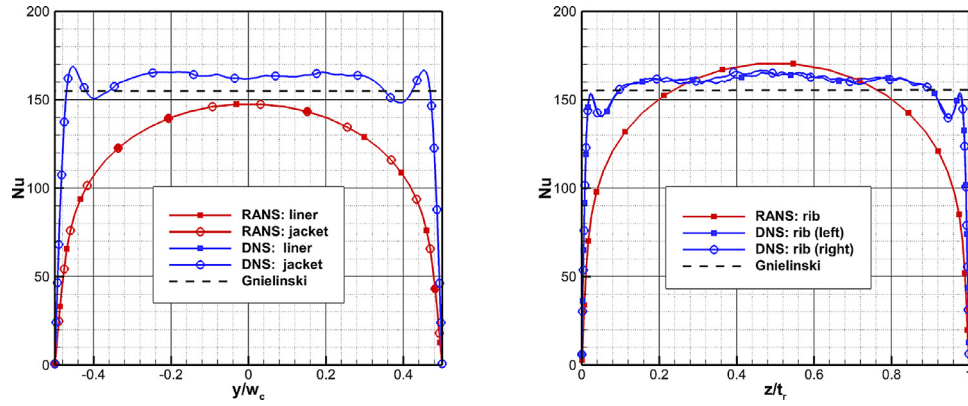


Fig. 13. ISOTH flow case: local Nusselt number distributions along the liner and jacket wall (left) and the rib walls (right).

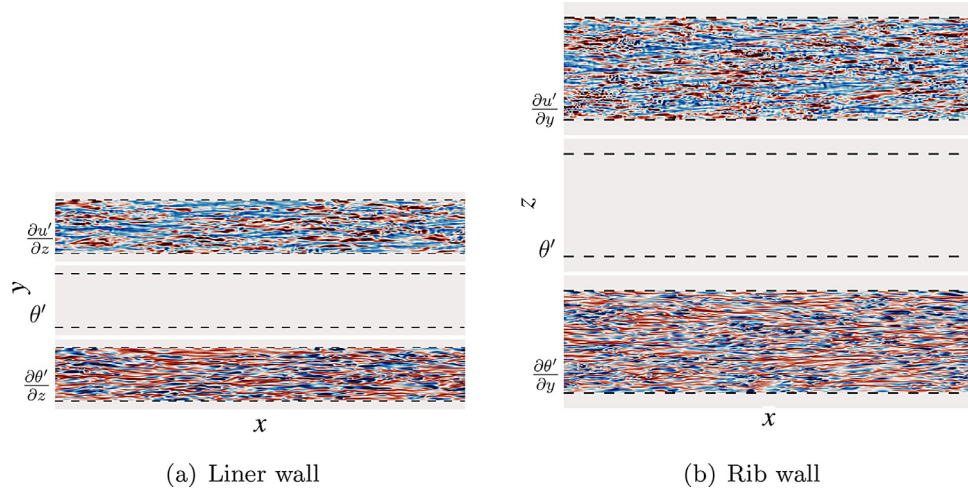


Fig. 14. DNS of ISOTH flow case: contours of local wall-normal velocity gradient, wall temperature, and wall-normal temperature gradient fluctuations at the liner wall (a), and at the rib wall (b). The color map is from blue (negative values) to red (positive values). The dashed lines denote the duct walls. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

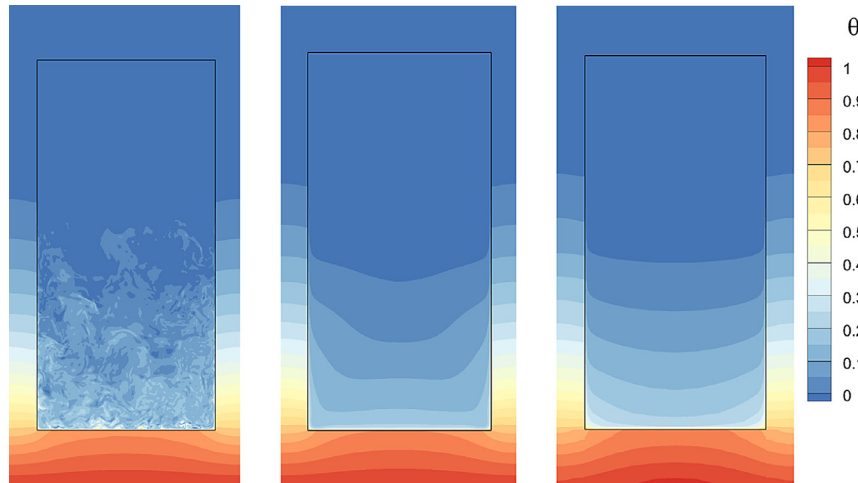


Fig. 15. HIGHCOND flow case: nondimensional temperature fields (θ) from DNS (instantaneous, left, and averaged, middle), and at channel exit from RANS/Fourier (right).

thermal diffusion, the temperature distribution may be strongly affected by thermal diffusion within the solid, as expressed by the ratio k_s/k_f . The temperature fields obtained with DNS and RANS/Fourier are shown for the high-conductivity case (HIGHCOND) in Fig. 15. Asymmetric heating yields a temperature field much different from that seen in the isothermal case, in line with the study of Straub et al. [25], who studied the flow in a circular

pipe with asymmetric heat load by DNS. In fact, the figure makes it clear that heat is mainly coming from the bottom, it is conducted through the wall, and then redistributed through the fluid by turbulent convection. Significant heat transfer also takes place within the rib walls, as one can judge from the temperature isolines. Temperature variations within the duct walls drive the temperature field in the fluid, which shows significant thermal strat-

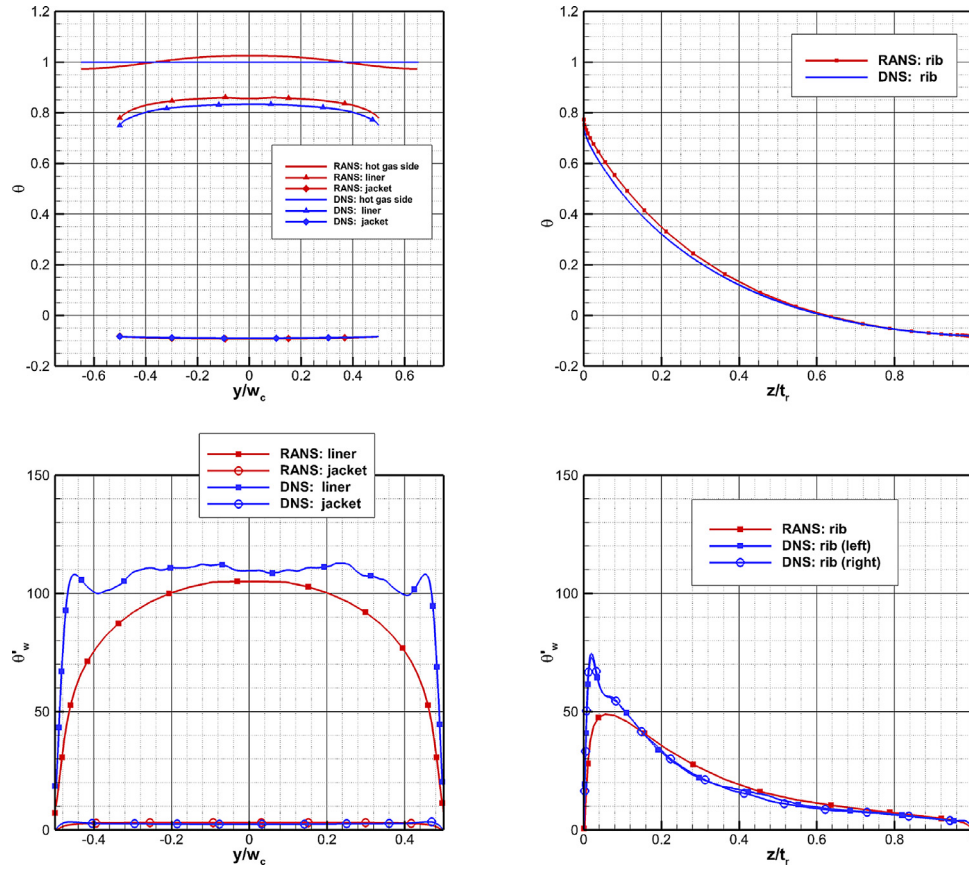


Fig. 16. HIGHCOND flow case: local temperature and wall-normal temperature gradient distributions along the liner and jacket wall (left) and along the rib walls (right).

ification controlled by the temperature distribution along the rib walls. In fact, it appears that the turbulent transport of heat is only active in the lower half of the duct, whereas temperature is nearly uniform in the upper half. Hence, the expectation is that the duct is less effective in redistributing heat than in the isothermal case.

Comparing the instantaneous and averaged DNS temperature field, it can also be recognized that the temperature field within the solid wall is barely affected by instantaneous fluctuations of temperature within the fluid. Thermal inertia of the solid wall is such as to damp all fluctuations, and differences between instantaneous and averaged temperature distributions within the walls are negligible. It is quite remarkable that RANS/Fourier in this case predicts a mean temperature field within the walls which is very close to the DNS estimates, despite visible differences within the fluid. In fact, temperature distribution within the duct retains the differences observed in the ISOTH case, in that RANS tends to smooth the distribution at the corners as compared to the DNS solution, which better follows the wall geometry owing to the presence of secondary motions. The result is that the secondary motion yields more efficient cooling of the corner regions, where the temperature predicted by DNS is lower than by RANS/Fourier. As a result, the temperature isolines seem to have opposite curvature near the liner wall.

Again, to quantify the temperature fields from an engineering point of view, we resort to a comparison of the heat transfer coefficient through the fluid Nusselt number, as reported in Table 3. As expected based on previous observations of nonuniform heating along the duct walls, the fluid Nusselt number is reduced as compared to the isothermal case, which is optimal in this respect. The Gnielinski correlation does not account for this effect, and in fact the existing literature [9,26] suggests that it may overestimate

the heat transfer by as much as 20% in the case of asymmetrically heated rectangular ducts, which is consistent with the difference herein found with respect to DNS data. The decrease of the Nusselt number with respect to the isothermal case is captured by RANS/Fourier, although it is somewhat overemphasized, with an underprediction of the heat transfer coefficient by 13%. The overall heat exchange property of the system fluid + solid is well represented by the global heat transfer coefficient, $\bar{Nu}_g$ as defined in Eq. (18). Differences are in this case less than for the fluid only, which deserves some explanation. In fact, the overall thermal resistance of the CHT system is the sum of the thermal resistances of the fluid and the solid, and since thermal diffusion within the solid is well characterized both in RANS/Fourier and DNS, the difference in the overall thermal resistance is less than for the fluid only.

The observed differences in the heat transfer properties can be analyzed in greater detail based on the local behavior of the wall temperature and wall-normal temperature gradient, shown in Fig. 16. Differences are mainly concentrated in the lower part of the cooling channel, where RANS/Fourier has generally higher temperatures than DNS, consistent with the lower thermal efficiency of the fluid. Specifically, the average temperature at the liner wall predicted by DNS is $\bar{\theta}_{w,l} = 0.816$, whereas RANS yields $\bar{\theta}_{w,l} = 0.843$. Notably, these values are consistent with the simple estimate given in Eq. (22), which yields $\bar{\theta}_{w,l} = 0.828$ for DNS, and $\bar{\theta}_{w,l} = 0.848$ for RANS, based on the respective fluid Nusselt numbers. Moving away from the hot side, the temperature distributions from RANS and DNS tend to coincide, and in fact the mean temperatures at the jacket wall are nearly identical and less than the bulk value in the channel, which implies a sign change of θ along the rib walls, at about two-thirds of the rib height. Despite the

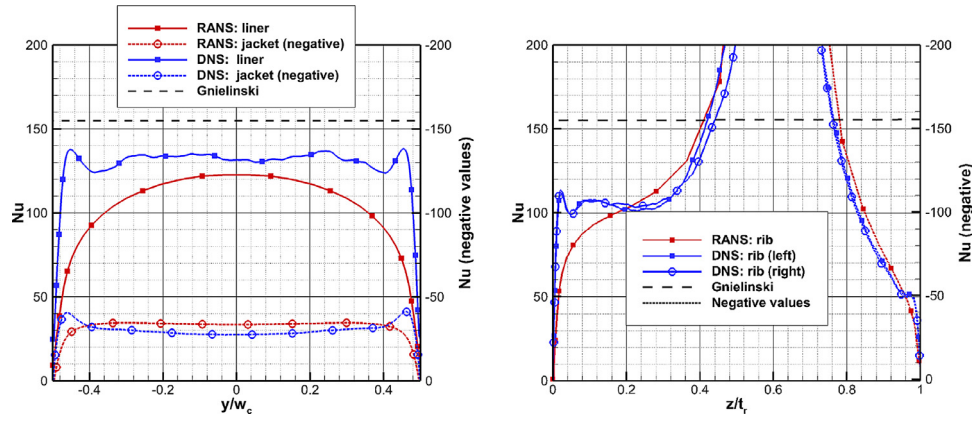


Fig. 17. HIGHCOND flow case: local Nusselt number distributions along the liner and jacket wall (left) and the rib walls (right).

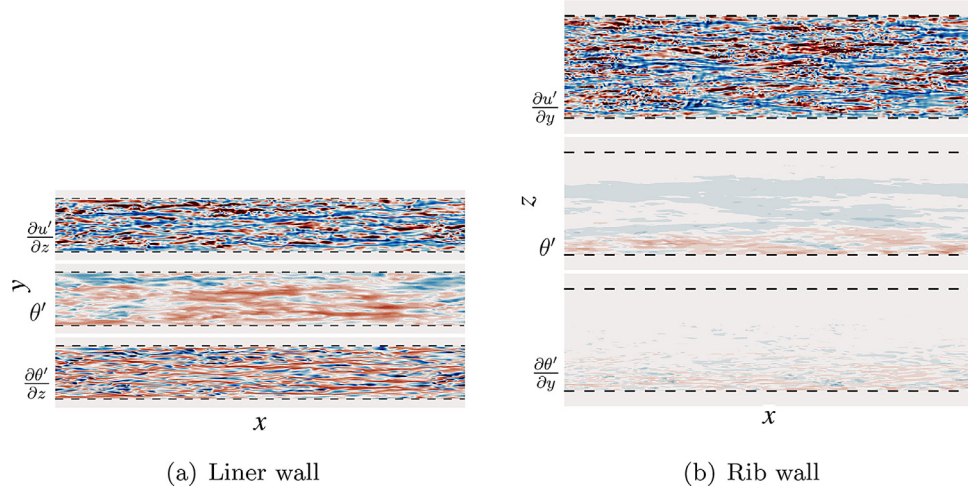


Fig. 18. DNS of HIGHCOND flow case: contours of local wall-normal velocity gradient, wall temperature, and wall-normal temperature gradient fluctuations at the liner wall (a), and at the rib wall (b). The color map is from blue (negative values) to red (positive values). The dashed lines denote the duct walls. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

sign inversion of the local temperature, the temperature gradient is not changing sign, and the heat flux is always towards the fluid, however, with large variations along the duct perimeter. Looking at the liner wall, and like in the ISOTH case, the distribution of the heat flux is parabolic in RANS/Fourier and flatter in DNS, with RANS/Fourier yielding lower heat exchange. Some difference is retained in the lower part of the ribs, whereas the distributions tend to collapse in the upper part of the ribs and along the jacket wall.

The resulting distributions of the local Nusselt number are shown in Fig. 17. Recalling the definition given in Eq. (20), a change of sign of θ along the rib walls implies a change of sign of Nu , which is accounted for in the graphs by using two axes, one for positive Nu and one for negative. Recalling again that the average value of the local Nu does not coincide with the mean Nusselt number, in the figure we nevertheless report as reference the value of $\overline{Nu}_f$ predicted by the Gnielinski correlation, that we have seen as a good reference value in the ISOTH case. The distributions of the local Nu along the liner wall well reflect the distributions of the wall-normal temperature gradient, and the associated differences between RANS/Fourier and DNS. Differences are much smaller in the upper part of the duct, where however, the Nusselt number is much less, reflecting a lower efficiency in the heat transfer process. This is a consequence of the fluid thermal stratification. It is evident from the velocity field that there is no difference in the convection process along the perimeter. However, heat exchange is related to the difference between the wall and

fluid temperature. If in the isothermal case a single fluid temperature acting on the heat exchange can be clearly identified as the bulk temperature, in case of thermal stratification the fluid temperature seen from the wall is different depending on the position along the perimeter. This is clear if one compares the heat transfer at the jacket and liner walls. As seen above, the heat flux is in both cases from the wall to the fluid, despite in the former case the wall temperature is lower and in the latter is higher than the bulk temperature. The role of bulk temperature therefore changes and becomes, in this case a useful mean to assess the reduced heat exchange performance induced by thermal stratification.

The correlation between temperature and velocity fluctuation is significantly affected by nonuniformity of the temperature distributions within the duct walls. Instantaneous fluctuating velocity and temperature fields obtained with DNS are shown in Fig. 18. Obviously, the limiting velocity field exhibits the same streaky structure as in the isothermal case. However, the behavior of the temperature field is quite different. At the liner wall (a), a negative correlation between temperature and velocity fluctuations is still visible, and it is especially clear that intense high-temperature events are preferentially found in connection with low-speed streaks. The main difference from the isothermal case appears to be the weaker intensity of the fluctuating thermal flux, which results from smaller temperature difference. It is also quite interesting to note that the fluctuating temperature field (middle panel) is closely correlated with the temperature gradient, how-

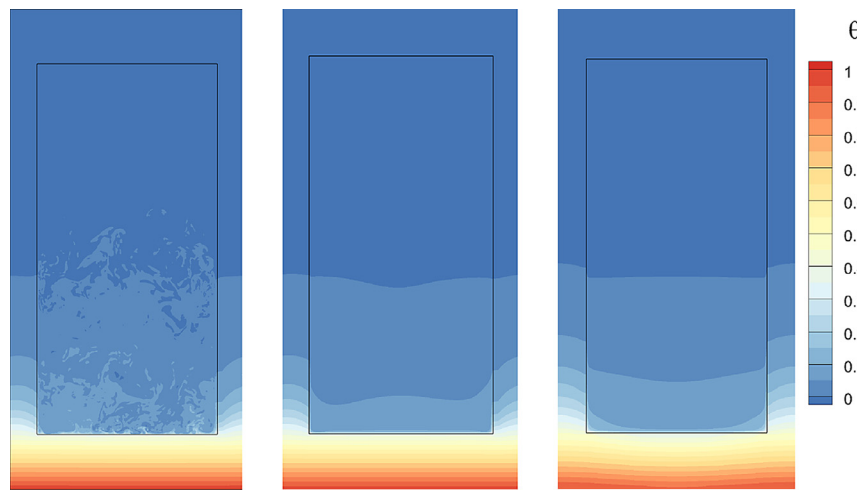


Fig. 19. LOWCOND flow case: nondimensional temperature fields (θ) from DNS (instantaneous, left, and averaged, middle), and at channel exit from RANS (right).

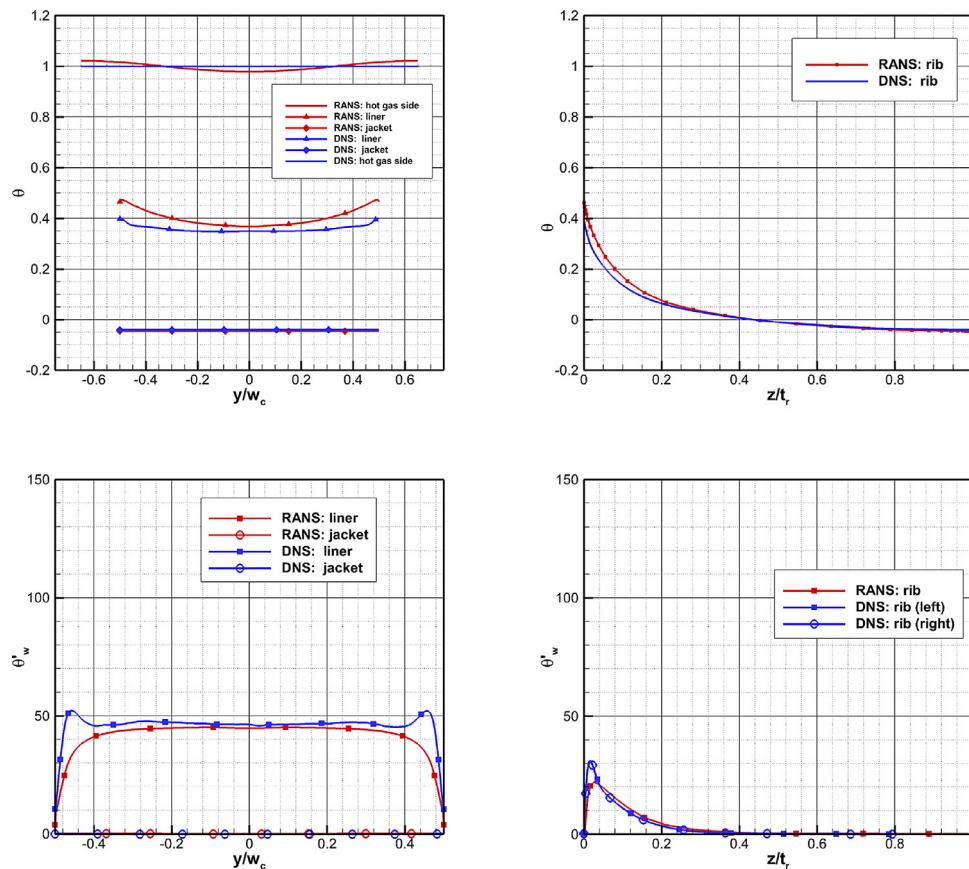


Fig. 20. LOWCOND flow case: local temperature and wall-normal temperature gradient distributions along the liner and jacket wall (left) and along the rib walls (right).

ever the small-scale fluctuations are filtered off, as a result of the higher thermal inertia of the material. At the rib wall (b) the association between velocity and temperature fluctuations is still detectable upon careful inspection, although heat flux and temperature fluctuations are much smaller than at the liner wall (for the sake of comparison, the same color scale is used).

5.4. Temperature field: low conductivity ratio

The LOWCOND test case is meant to bring out the effects of further reduction of the thermal conductivity of the solid walls. The nondimensional temperature θ flow field clearly shows clear temperature fluctuations within the fluid phase superposed to a well-

defined average distribution (see Fig. 19). Large temperature gradients now take place within the liner wall, as a consequence of the low thermal conductivity of the solid. Contrary to the HIGHCOND case, the temperature isocontours tend to be nearly parallel within the liner wall, and temperature increase due to conduction within the rib walls is less significant. Hence, temperatures higher than the bulk are confined to the bottom of the channel, whereas the upper part of the system is not taking any significant role in the heat redistribution process, and temperature fluctuations within the fluid are also very small. As in the HIGHCOND case, the effects of temperature fluctuations within the fluid are again damped by large thermal inertia of the solid walls, which instead have a remarkably steady temperature distribution. Also in

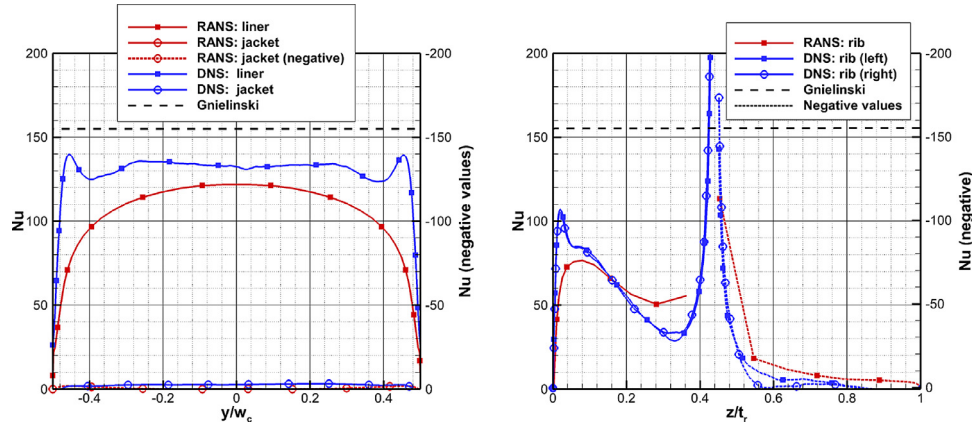


Fig. 21. LOWCOND flow case: local Nusselt number comparison along the liner and jacket wall (left) and the rib wall (right).

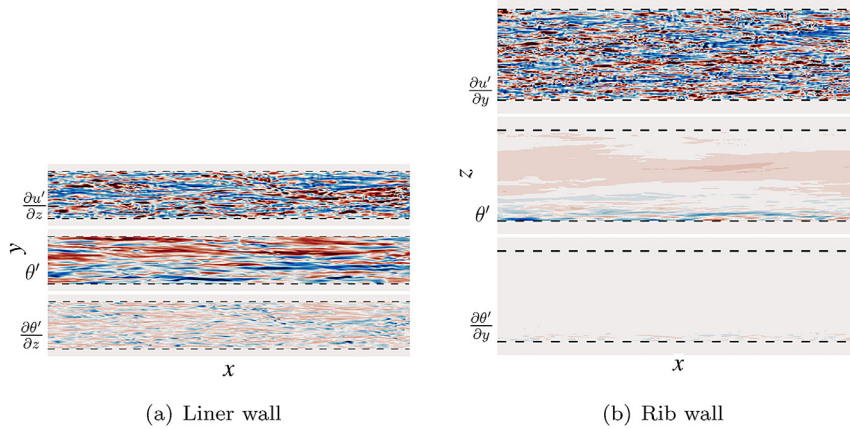


Fig. 22. DNS of LOWCOND flow case: contours of local wall-normal velocity gradient, wall temperature, and wall-normal temperature gradient fluctuations at the liner wall (a), and at the rib wall (b). The color map is from blue (negative values) to red (positive values). The dashed lines denote the duct walls. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

this case the presence of secondary motions in the DNS manifests itself through inversion of the curvature of the temperature isolines near the liner wall.

Comparison of the temperature and wall-normal temperature gradients along the duct walls is shown in Fig. 20. As in the HIGHCOND case, the main differences in the temperature distributions are found in the lower part of the system. In particular, the mean temperature at the liner wall is higher in RANS/Fourier ($\bar{\theta}_{w,l} = 0.401$) than in DNS ($\bar{\theta}_{w,l} = 0.357$), with a difference of about 12%. The prediction of the approximate formula (22) is particularly good for this case, being $\bar{\theta}_{w,l} = 0.402$ for RANS/Fourier, and $\bar{\theta}_{w,l} = 0.335$ for DNS, which is a quantitative demonstration that heat transfer through the rib walls has minor importance in this case. Differences between RANS/Fourier and DNS temperature distributions persist up to 40% of the rib height, at which that the bulk temperature is reached. Past this distance, RANS/Fourier predictions coincide with DNS. Heat flux across the liner wall shows similar distributions as in the HIGHCOND case, although the temperature gradients are lower in the present case. Heat transfer through the upper part of the rib walls and through the jacket wall is entirely negligible.

The above noted differences are reflected in the distributions of the local Nusselt number, shown in Fig. 21. Quite interestingly, the behavior of Nu along the liner wall is nearly identical to the HIGHCOND case, for both RANS/Fourier and DNS, which is probably the result of universality in the presence of a single heated wall. The Nusselt number distributions along the rib walls again highlight

underestimation of heat transfer near the lower corners and show the presence of a singularity corresponding to the sign inversion of θ_w .

All the above information contributes to interpretation of the average fluid Nusselt number ($\bar{Nu}_c$), which is reported in Table 3. In the LOWCOND case, we find that $\bar{Nu}_c$ is further reduced as compared to the isothermal case, which we may interpret as a result of ineffectiveness of most walls in the heat transfer process. RANS/Fourier seems to be more sensitive than DNS to this change. The reason, referring to the ISOTH results, might reside in the larger relative importance of the liner wall, where the discrepancy with respect to DNS is higher than for the longer rib walls. The global performance of the CHT system is significantly degraded by the decreased thermal conductivity of the solid, as Table 3 clearly shows. However, the differences between RANS/Fourier and DNS are attenuated by greater importance of thermal conduction within the solid, which is well captured by the RANS/Fourier solver.

Instantaneous fluctuating velocity and temperature fields found with DNS for the low-conductivity case are shown in Fig. 22. Comparison with the high-conductivity case at the liner wall (a) shows a similar qualitative behavior, however the fluctuating heat flux is even less in the present case. The in-plane temperature fluctuations are again 'wider' than the heat flux fluctuations, but the smearing effect is less than in the high-conductivity case (compare with Fig. 18(a), as the solid thermal inertia is now smaller. At the rib wall (b) we again observe much weaker temperature fluctuations on account of the reduced mean temperature gradients.

6. Discussion and conclusions

Continuous technological innovations in space propulsion and particularly in liquid rocket engines, require adequate and innovative means to provide support in the development phase. In this paper we consider the prototype case of heat exchanger consisting of a rectangular channel, under the simplifying assumption of uniform and isotropic properties of fluid and solid media. Two conjugate heat transfer approaches have been compared, one based on the coupled solution of the RANS and Fourier equations for the fluid and solid domains, and the second based on seamless DNS of the Navier-Stokes equations and of the solid domain. The RANS closure selected in the present study is based on the Spalart-Allmaras one-equation model because of its simplicity and widespread use. As a consequence, the discussion is strictly related to this model, while specific simulations should be carried out to discuss the behavior of different RANS closures. The results herein presented for conjugate heat transfer with different values of the solid-to-fluid conductivity ratio unveil a number of important issues, and pinpoint discrepancies of correlations of standard use, as well as some difference between RANS and DNS results. First of all, we find that the discrepancy of reference correlations with respect to DNS data is within expectations in the ISOTH case (infinite solid conductivity). According to the literature, errors within $\pm 9\%$ are expected for non-circular ducts with uniform heating [10]. In fact, we find that difference of the fluid Nusselt number estimated by the Gnielinski analogy with respect to DNS is negligible in the ISOTH case. Deviations are larger when finite solid conductivity is accounted for. In general, the effect of finite solid conductivity is a strongly asymmetric, yet distributed, heat load on the fluid, as most heat transfer becomes concentrated at the liner wall. In this respect, we note that the study of rectangular ducts under asymmetric heat load conditions has been rarely addressed in the past, despite its great practical relevance [27,28]. No general conclusion seems to arise from past studies, and in fact the general recommendation which is given in heat transfer handbooks is to use the same estimates as for uniform heating [10]. Here, both DNS and RANS consistently suggest that in the presence of finite wall conductivity the global fluid Nusselt number decreases by decreasing the solid conductivity. Specifically, reductions with respect to the ISOTH case amount to about 12% in DNS, and 17% in RANS, which is quite substantial.

Decrease of the thermal efficiency of the fluid as a result of finite conductivity within the solid is here explained as a result of the thermal stratification induced by the finite wall thermal conductivity, whereby only the lower part of the channel is taking part in the convective heat transfer, whereas mean temperature gradients in the upper part of the duct are very small. This effect is found to be monotonic with the k_f/k_s parameter. Consistent with general claims [27], we find that the Nusselt number at the most heated (liner) wall decreases as compared to the isothermal case, however the values predicted by DNS and RANS/Fourier are (see Figs. 17 and 21) $\bar{Nu}_l = 128$ and $\bar{Nu}_l = 104$, respectively, quite far from the only known specific correlation for rectangular ducts with heating on one side only [28], which instead would yield an unrealistically high $\bar{Nu}_l = 176$.

Despite inherent limitation of the RANS model used here, which relies on linear eddy-viscosity parametrization of the turbulent shear stress, hence precluding the appearance of secondary eddies, we find that RANS does a good job in predicting pressure losses within the duct, with $O(1\%)$ error. The main qualitative difference with respect to DNS is a more rounded distribution of the friction and heat transfer coefficients, which are nearly flat in the DNS. Differences in the heat transfer performance of the duct are larger than for friction, from 10% in the ISOTH case to 17% in the LOWCOND case. Discrepancies between RANS and DNS in predic-

tion of the fluid Nusselt number translate into differences in the prediction of the mean liner wall temperature, which is higher in RANS/Fourier. Increasing accuracy of the correlation (22) in predicting the liner wall temperature is a symptom of reduced importance of thermal diffusion through the rib walls, which is in fact supported by analysis of the temperature fields.

The most important practical information in CHT is the global heat transfer coefficient of the system $\bar{Nu}_g$ as defined in Eq. (18), which expresses in non-dimensional form the ratio between the imposed heat flux through the bottom of the liner wall and the resulting temperature difference with respect to the fluid bulk. Equation (19) suggests that the overall performance of the CHT system is proportional to the fluid Nusselt number ($\bar{Nu}_c$) and to the ratio of the difference between the mean duct wall temperature and bulk temperature, and the total temperature difference (expressed by $\bar{\theta}_{w,c}$). This expression suggests a two-fold effect of conjugate heat transfer, which indirectly affects $\bar{Nu}_c$ as discussed below, and also directly affects $\bar{\theta}_{w,c}$. In this sense, the ISOTH case is optimal, as it yields both optimal $\bar{Nu}_c$ owing to uniform heating conditions, and maximum $\bar{\theta}_{w,c} = 1$. Finite conductivity of the solid walls yields mild decrease of $\bar{Nu}_c$, as well as large decrease of the mean peripheral temperature along the duct walls, resulting in large decrease of $\bar{Nu}_g$, which is well visible in Table 3. When all the effects are lumped together, our model based on a coupled RANS/Fourier approach does an overall good job, with a maximum error of about 10% in the ISOTH case, reducing to less than 5% in the LOWCOND case as a result of the greater relative importance of conductive heat transfer as compared to the convective one.

We believe that the data reported in this paper may contribute to the development of advanced prediction methods for cooling channels of rocket engines, which should be capable of accurately accounting for the finite thermal conductivity of the walls. A few final comments and suggestions for future work should be issued. First, we believe that there is room for improvement for RANS models. Here, a simple Spalart-Allmaras model was tested, but it would be worth verifying the accuracy of more elaborate models based on full Reynolds stress closures. In fact, we expect that full Reynolds stress models can correctly predict the presence of secondary motions, with incurred benefits in terms of realism in the turbulent transport of momentum and heat. Second, we believe that classical formulas based on the assumption of uniform heating should be used with great caution in CHT problems, as coupling with heat conduction in the solid phase yields large non-uniformity which in turn imply reduction of the overall system cooling performance. Last, we believe that additional computational and experimental studies should be carried out to achieve a better characterization of cooling channels (here, of rectangular shape) as a function of non-symmetric heat loading conditions, as existing correlations do not provide sufficient predictive accuracy.

Declaration of Competing Interest

None.

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